

EDA & Data Cleaning Summary Report

Dataset: Cars93 (1993 car models)

1. Dataset Overview

The dataset used for this project is **Cars93**, a well-known dataset of 93 car models from the 1993 model year, containing 27 attributes covering pricing, engine specifications, fuel efficiency, dimensions, and passenger capacity. This version of the dataset has intentionally-introduced missing values, making it a good candidate for practicing data cleaning techniques.

Property	Value
Rows (cars)	93
Columns (attributes)	27
Total missing cells (raw)	170
Duplicate rows	0
Missing cells after cleaning	0

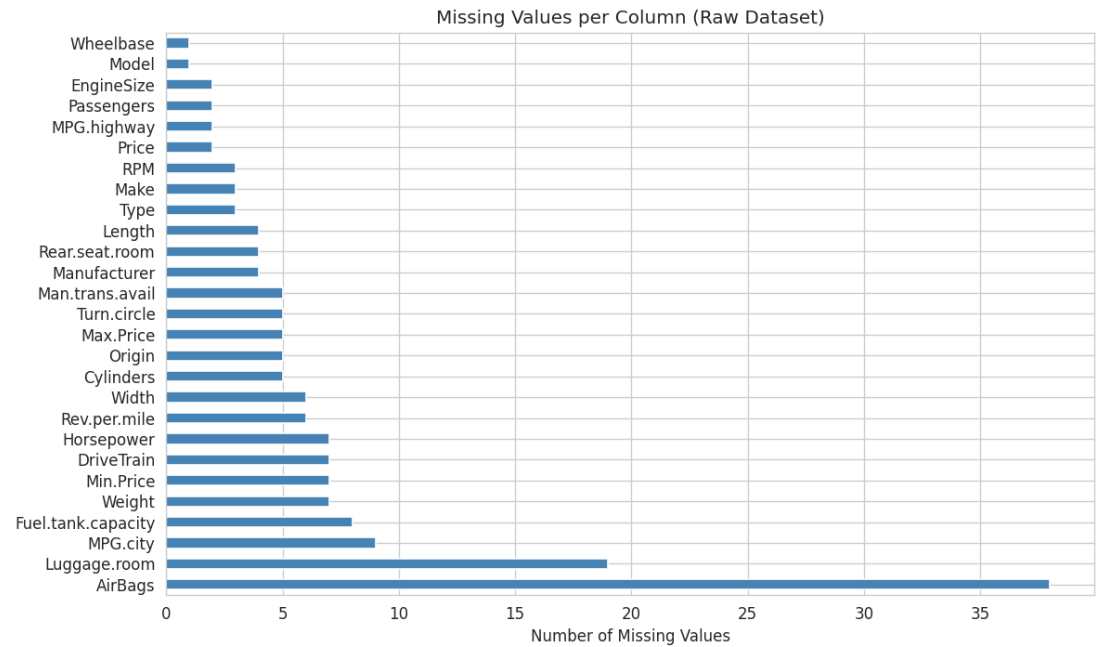


Figure 1: Count of missing values per column in the raw dataset.

2. Cleaning Steps

The following cleaning steps were applied, in order, using Pandas:

- **Duplicate check:** Checked for exact duplicate rows using `df.duplicated()` — none were found.
- **Numeric columns:** Missing values (e.g. Price, Horsepower, Weight) were filled with the column **median**, which is robust to outliers and skewed distributions.
- **Categorical columns:** Missing values (e.g. Manufacturer, AirBags, DriveTrain) were filled with the column **mode** (most frequent value).
- **Text formatting:** Leading/trailing whitespace was stripped from all text columns for consistency.
- **Export:** The cleaned dataset was saved to `data/cleaned_dataset.csv`.

After cleaning, the dataset has **zero missing values** across all 27 columns, with the original 93 rows preserved.

3. Key Insights

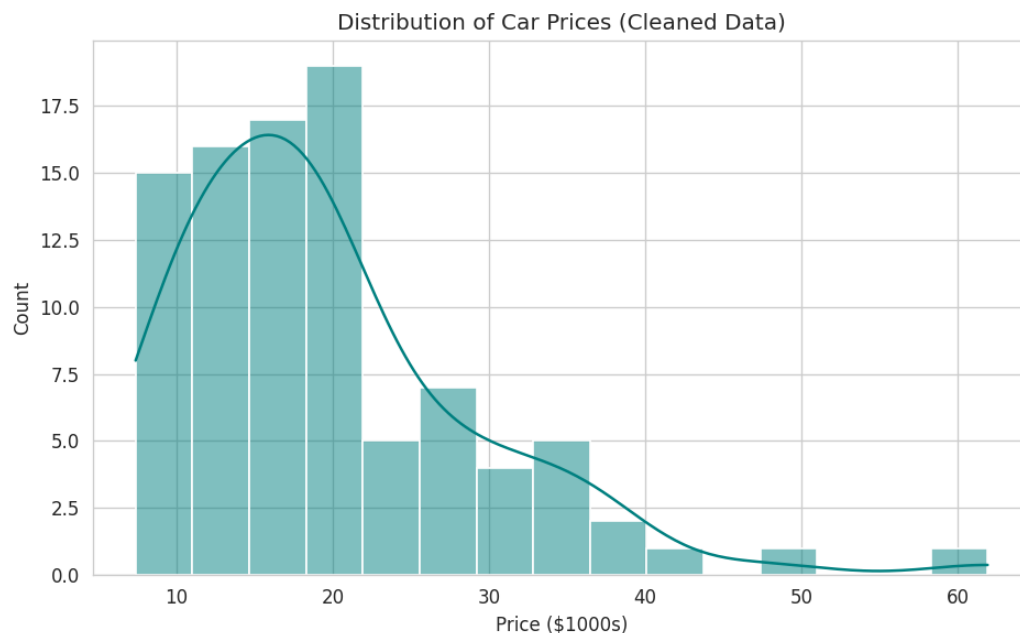


Figure 2: Distribution of car prices — right-skewed, with most cars in the \$10k-\$25k range and a smaller number of higher-end outliers.



Figure 3: Price by car type — Sporty and Midsize categories show the widest price ranges and the most high-end outliers.

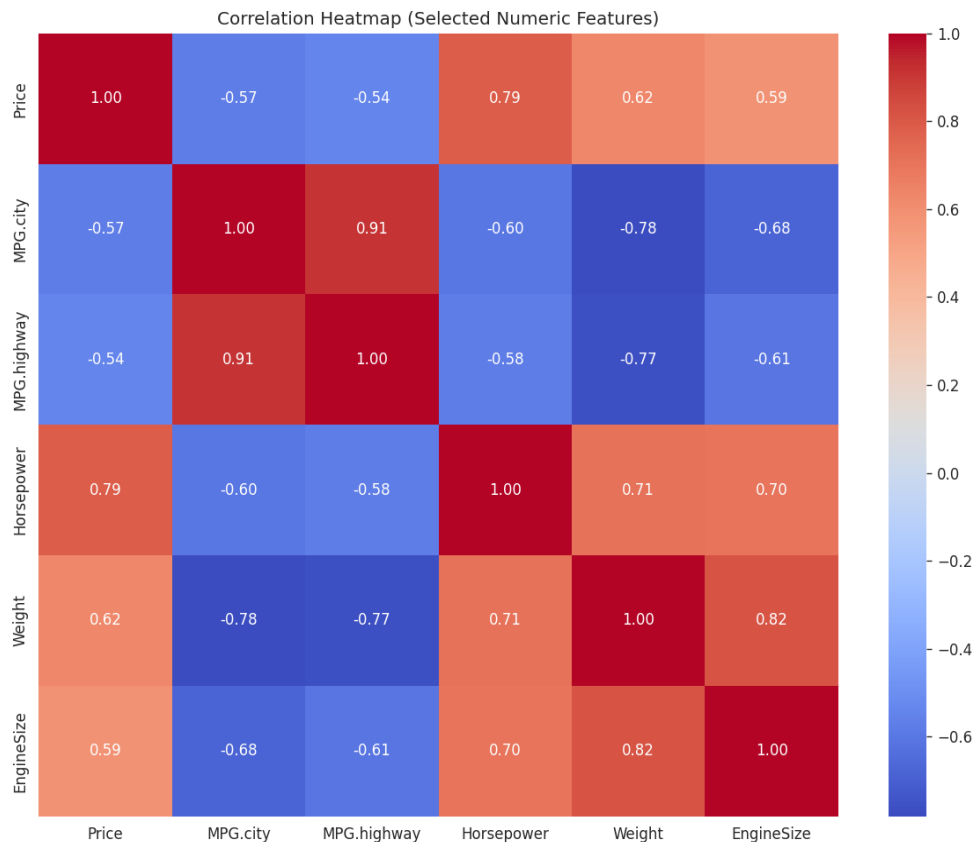


Figure 4: Correlation between price and key numeric features. Weight and Horsepower are positively correlated with Price, while MPG (city and highway) are negatively correlated with Price.

- Car prices are right-skewed: most cars are moderately priced, with a small number of high-end outliers.
- Heavier, more powerful cars tend to cost more and be less fuel-efficient (negative correlation between Price and both MPG.city and MPG.highway).
- AirBags and Luggage.room had the highest proportion of missing data among all columns (41% and 20% respectively), suggesting these fields may have been optional or inconsistently recorded for some models.

4. Challenges Faced

- **High missingness in some columns:** AirBags was missing for 41% of rows, which made simple mode-imputation less reliable for that specific column compared to others with lower missingness.
- **Mixed data types:** The dataset mixes numeric (Price, Weight) and categorical (Manufacturer, Type, DriveTrain) columns, requiring separate imputation strategies for each type.
- **Small sample size:** With only 93 rows, imputed values (median/mode) can have a proportionally larger effect on overall statistics compared to a larger dataset.

Full analysis code, including all cleaning steps and visualizations, is available in notebooks/eda_and_cleaning.ipynb.